

Anish Kumar

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Education

DY Patil International University, Akurdi, Pune

Aug 2024 – May 2028

- Bachelor of Engineering (Computer Science and Engineering) | SGPA: 8.85/10.0
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks

Projects

StreamForge — Video Streaming Platform | Node.js · React · FFmpeg · PostgreSQL · Redis ·

Jan 2026

BullMQ · MinIO

GitHub | Live

- Built an HLS transcoding pipeline using FFmpeg to generate adaptive 360p/720p streams on MinIO (S3-compatible); benchmarked sub-3s stream start time on uploads averaging 500 MB
- Designed an event-driven upload pipeline with presigned S3 URLs, BullMQ background workers (concurrency: 2), and Socket.IO for real-time progress — handled 20+ concurrent uploads without queue starvation
- Implemented a hybrid recommendation engine combining trending scores, personalized feeds, collaborative filtering, and co-watch signals; chose a 40/40/20 weighting to balance discovery with relevance for new vs. returning users
- Instrumented the backend with prom-client exposing a /metrics Prometheus endpoint; built full-text search with tsvector and pg_trgm with Redis-cached ranking — reduced query response time to under 80 ms on a 10,000-row dataset

LinkHub — Link-in-Bio SaaS Platform | Node.js · React · PostgreSQL · Stripe · Docker

Feb 2026

GitHub

- Engineered a three-tier freemium system (Free/Pro/Premium) with Stripe Checkout, monthly/annual billing, and cancel/resume flows; built a ProGate feature-gate using React.cloneElement to intercept clicks without modifying child components
- Built real-time mobile preview with Zustand subscriptions, a 7-section design panel, and a wallpaper engine supporting gradients, image effects, patterns, and video backgrounds
- Implemented Google OAuth (Passport.js) + JWT auth with per-user Zustand localStorage persistence rehydrated on login; built analytics tracking profile views and per-link clicks with geo and device metadata
- Containerised the full stack with Docker (React/NGinx frontend, Node.js backend, PostgreSQL) with long-lived cache headers (Cache-Control: max-age=31536000) and React Router fallback support

Nexus — AI-Powered Search Engine | Node.js · React · Google Gemini · Supabase · Tavily

Mar 2026

GitHub | Live

- Built a Perplexity-style AI search engine: fetches real-time results via Tavily API, passes them as grounded context to Google Gemini, and streams cited answers via SSE — reducing hallucination vs. a vanilla LLM call
- Designed streaming response architecture achieving perceived first-token latency under 400 ms; deployed frontend on Vercel and backend on Render with environment-specific Supabase OAuth callback configuration
- Implemented full-stack auth with Supabase (GitHub and Google OAuth), configuring redirect URLs and provider callbacks for both local and production environments
- Structured the backend with a domain-driven src/modules/ layout separating search, auth, and streaming concerns; maintained clean dev/production parity across Supabase projects and environment configs

Technical Skills

Languages: JavaScript, TypeScript, Python, C++

Frontend: React.js, Next.js, Tailwind CSS, GSAP, Zustand

Backend: Node.js, Express.js, BullMQ, Socket.IO, WebSockets, WebRTC, Passport.js, Prisma ORM

Databases & Storage: PostgreSQL, Redis, MongoDB, MinIO, Supabase

Tools & DevOps: Git, Docker, FFmpeg, Prometheus, Grafana, Stripe, Render, Vercel, Upstash

Leadership & Achievements

- FullStack Lead — Code-Decode Club (Official Coding Club, DY Patil International University, 2024–Present)
- Full-Stack Bootcamp Instructor — Designed and led a 10-day intensive bootcamp for 80+ students covering React, Node.js, PostgreSQL, and JWT auth; created lesson plans, live-coding sessions, and capstone briefs
- Hackathon Magnitude 2.0 — Built an AI-powered crop management platform with Google Gemini-based pest detection and real-time irrigation monitoring; competed against teams from multiple institutions